

Problem Statements for CUPGS Internal Hackathon

1. Development of an Innovative Flow-Control Mechanism for Improving the Performance of Small-Scale Wind Turbines

Small-scale wind turbines often suffer from poor aerodynamic performance under low and highly variable wind-speed conditions. Conventional blade geometries may not provide satisfactory starting torque and aerodynamic efficiency across a wide range of operating conditions.

Challenge

Develop a novel passive or semi-active flow-control mechanism that can be integrated with a small-scale wind-turbine blade to:

- improve starting capability,
- enhance aerodynamic performance,
- delay flow separation,
- improve torque/power generation under low-speed conditions, and
- maintain acceptable performance over varying wind speeds.

The solution should preferably be simple, manufacturable, and retrofittable to an existing small wind turbine.

Expected output

- Conceptual/CAD design
- Prototype or scaled model
- CFD and/or experimental validation
- Performance comparison with a conventional blade
- Preliminary assessment of patentability

2. Development of an Innovative Modular Water-Tunnel Test Section for Enhanced Flow Quality and Experimental Flexibility

Experimental investigation of flow around bluff bodies, hydrofoils and other submerged geometries requires a water tunnel with good flow uniformity, low turbulence, easy model installation and rapid modification of the test section. Conventional laboratory arrangements may require considerable time for changing test models and experimental configurations.

Challenge

Develop a novel modular water-tunnel test-section or model-mounting arrangement that enables:

- rapid installation and removal of test models,
- accurate positioning of models,
- minimal disturbance to the upstream flow,
- improved flow uniformity,
- easy adaptation for different experimental configurations, and
- convenient integration with flow-visualization or measurement systems.

The proposed arrangement should be suitable for laboratory-scale experimental investigations in fluid mechanics and aerodynamics/hydrodynamics.

Expected output

- CAD model
- Working prototype/mock-up
- Flow visualization
- Experimental validation
- Comparison with conventional arrangements
- Assessment of scalability and patent potential

3. Smart Irrigation & Soil Moisture Monitoring

Problem Title: AI-Based Smart Irrigation System Using Soil Moisture and Weather Data

Problem:

Farmers often irrigate crops based on fixed schedules rather than the actual moisture condition of the soil. This can lead to over-irrigation, water wastage, nutrient loss, and poor crop productivity. Develop a low-cost IoT-based system that continuously measures soil moisture at different depths and uses weather conditions to determine when and how much water should be supplied.

Expected solution:

- Soil-moisture sensors at multiple depths.
- Temperature, humidity and rainfall sensing.
- Weather forecast integration.
- AI/ML model to predict soil moisture and irrigation requirement.
- Automatic control of pump/solenoid valves.
- Mobile/web dashboard showing moisture status and water consumption.
- Alerts when soil becomes excessively dry or irrigation is unnecessary.

Possible innovation: Predict the moisture level for the next 6–24 hours and irrigate before the soil reaches the critical moisture level.

4. Intelligent Transportation & Traffic Management at Busy Junctions

Problem Title: AI-Based Adaptive Traffic Signal and Congestion Management System for Busy Urban Junctions

Problem:

Conventional traffic signals operate using fixed timings even when traffic volume varies considerably between approaches. This causes unnecessary waiting, congestion, fuel consumption and increased emissions. Develop an intelligent traffic management system that uses real-time traffic information to dynamically control signal timings at a busy junction.

Expected solution:

- CCTV/camera or low-cost sensors to estimate vehicle density.
- Vehicle counting and classification using computer vision.
- Detection of queues and congestion.
- Dynamic adjustment of green/red signal duration.
- Priority management for ambulances and fire vehicles.
- Pedestrian crossing detection.
- Prediction of congestion using historical traffic data.
- Dashboard showing traffic density, waiting time and junction performance.

Possible innovation: Instead of optimizing only one junction, develop a coordinated system for 3–5 consecutive junctions, where signals communicate with each other to create a “green corridor” for major traffic flows.

5. IOT-Based device for face scanning and recording attendance of students and faculties

Manual attendance-taking in educational institutions is time-consuming, prone to errors, and vulnerable to proxy attendance. Existing RFID or biometric fingerprint systems often require physical contact and can be slow during peak entry times. There is a need for a contactless, automated system that can reliably identify individuals and log attendance in real time.

Challenge

Develop an IoT-based face scanning attendance device that enables:

- accurate real-time face detection and recognition of students and faculty,
- rapid, contactless attendance marking with minimal queuing time,
- reliable performance under varying lighting and positioning conditions,
- secure storage and transmission of attendance data to a central database/server,
- easy enrollment and management of new users (adding/removing faces from the system), and
- convenient integration with existing institutional management systems (ERP, student databases, notification systems).

The proposed device should be suitable for deployment at entry points of classrooms, departments, or institutional gates for automated attendance tracking.

Expected output

- Face recognition software module (detection + recognition pipeline)
- Attendance database/dashboard for records and reports
- System testing under real classroom/institutional conditions
- Experimental validation (accuracy, recognition speed, false acceptance/rejection rates)
- Comparison with conventional attendance methods (manual, RFID, biometric fingerprint)
- Assessment of scalability and patent potential
- Cost Comparison and business model

6. IoT and AI/ML-Based Device for Detecting Open or Submerged Manholes During Rainy Season

During heavy rainfall, waterlogging often submerges or conceals open manholes, creating serious safety hazards for pedestrians and vehicles. Manual inspection and traditional signage are inadequate for timely detection and warning under such conditions. There is a need for an automated system that can detect uncovered, damaged, or hidden manholes in real time and alert relevant authorities or the public.

Challenge

Develop an IoT-based AI/ML manhole detection device that enables:

- accurate detection of open, missing, or damaged manhole covers under waterlogged/rainy conditions,
- reliable sensing despite poor visibility, turbid water, and submerged surfaces,
- real-time AI/ML-based classification to distinguish manholes from other road anomalies,
- timely alert generation to municipal authorities and/or the public,
- robust and weatherproof hardware design suitable for outdoor, flood-prone environments, and
- convenient integration with municipal monitoring systems, GPS tagging, and city safety infrastructure.

The proposed device should be suitable for deployment on roadways, drainage-prone areas, or municipal vehicles for real-time manhole hazard detection during monsoon conditions.

Expected output

- Hardware/device prototype (sensor unit, microcontroller, weatherproof housing)
- AI/ML detection model (trained for manhole/hazard identification)
- Field testing under simulated/actual rainy and waterlogged conditions
- Experimental validation (detection accuracy, false alarm rate, response time)
- Comparison with conventional methods (manual inspection, static signage, barricades)
- Assessment of scalability and patent potential
- Cost Comparison and business model

7.Preventive Industrial Motor Health Monitor: Vibration and Thermal Sensing for Predictive Failure Detection in Three-Phase Induction Motors

Three-phase induction motors are critical components in industrial automation systems, and unexpected failures can lead to costly downtime and production losses. Conventional maintenance schedules are often either too conservative, leading to unnecessary servicing, or too reactive, allowing failures to occur before intervention. There is a need for a low-cost, real-time monitoring solution that can detect early signs of motor degradation.

Challenge

Develop a preventive motor health monitoring system that enables:

- continuous acquisition of vibration and thermal data from an operating motor,
 - microcontroller-based processing to identify abnormal patterns indicative of impending failure,
 - early warning/alerting before a fault becomes critical,
 - minimal intrusion into existing motor installations,
 - reliable operation in industrial environments (noise, heat, electrical interference), and
 - scope for integration with broader plant IoT/monitoring networks.
- The proposed system should be suitable for predictive maintenance of three-phase induction motors in industrial automation settings.

Expected output

- Hardware/device prototype (sensor module, microcontroller unit, mounting arrangement)
- Signal processing/ML algorithm for fault pattern detection
- Real-time monitoring dashboard/alert system
- Testing on operating motors under varying load and fault-simulated conditions
- Experimental validation (detection accuracy, lead time before failure, false alarm rate)
- Comparison with conventional maintenance approaches (scheduled/reactive maintenance)
- Assessment of scalability and patent potential

8.Smart Substation Load Shedding Unit: Automated Microcontroller-Based Relay System for Peak Demand Management

Electrical substations often face periods of peak power demand that can strain the grid and risk instability or outages. Manual load-shedding practices are slow to respond and may not prioritize critical versus non-critical loads effectively. An automated, intelligent load-shedding mechanism can improve grid reliability while protecting essential operations.

Challenge

Develop an automated load shedding unit that enables:

- real-time monitoring of substation load and demand levels,
 - microcontroller-based decision-making to identify and disconnect non-critical loads during peak demand,
 - rapid and reliable relay switching with minimal disruption to critical loads,
 - accurate prioritization between critical and non-critical circuits,
 - easy reconfiguration of load priorities for different substation setups, and
 - convenient integration with existing substation monitoring/control infrastructure.
- The proposed system should be suitable for automated peak-demand management in industrial and utility substations.

Expected output

- Hardware/device prototype (microcontroller-relay unit, load-sensing circuit)
- Control logic/firmware for load prioritization and shedding decisions
- Testing on simulated multi-load substation setup
- Experimental validation (response time, switching accuracy, load prioritization correctness)
- Comparison with conventional arrangements (manual load shedding, fixed relay systems)
- Assessment of scalability and patent potential

9.Underground Cable Fault Distance Locator: Impedance-Based Circuit for Precise Fault Distance Estimation

Underground power cables are prone to faults that are difficult to locate due to their inaccessibility, and manual fault-tracing methods can be time-consuming and require extensive excavation. An accurate, non-invasive method to pinpoint the distance to a cable break can significantly reduce repair time and cost.

Challenge

Develop an underground cable fault distance locator that enables:

- accurate impedance-based measurement to detect the presence of a cable fault,
- precise calculation of the distance (in meters) to the fault location,
- minimal need for physical excavation or invasive testing,
- reliable operation across different cable types and lengths,
- ease of use for field technicians with simple setup and readout, and
- convenient integration with utility maintenance and monitoring workflows.

The proposed system should be suitable for fault diagnosis and maintenance of underground power distribution cables.

Expected output

- Hardware/device prototype (impedance-measurement circuit, display/readout unit)
- Fault-distance calculation algorithm/firmware
- Testing on cable segments with simulated fault conditions at known distances
- Experimental validation (distance measurement accuracy/error margin)
- Comparison with conventional methods (manual tracing, existing fault locators)
- Assessment of scalability and patent potential